

# Hamza Ahmad

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## SUMMARY

Final-year Computer Science student at the University of Peshawar with hands-on experience building end-to-end AI systems. Specializing in machine learning, Retrieval-Augmented Generation (RAG), agentic AI pipelines, and model deployment. Completed a full-stack AI capstone project integrating ML-based disease prediction with an LLM-powered medical chatbot. Seeking AI/ML internship or research opportunities to contribute to real-world intelligent systems.

## TECHNICAL SKILLS

**Machine Learning:** Scikit-learn, Logistic Regression, Random Forest, SVM, model evaluation (ROC-AUC, cross-validation), class imbalance handling (SMOTE)

**Deep Learning & NLP:** Neural networks, HuggingFace Transformers, sentence-transformers, embeddings

**RAG & Agentic AI:** LangChain, ChromaDB, Ollama (Llama 3.2), agentic tool routing, document ingestion pipelines

**APIs & Deployment:** FastAPI, REST API design, model serialization (Pickle), multi-service backend orchestration

**Data:** Pandas, NumPy, Matplotlib, Seaborn, data cleaning, feature engineering, UCI datasets

**Languages:** Python, Dart (Flutter), HTML, CSS, JavaScript

**Tools:** Git, VS Code, Android Studio, Streamlit, Jupyter Notebook

## FEATURED PROJECT

### AI-Driven Cardiovascular Disease Risk Prediction System

2025 – 2026

- Built an end-to-end CVD risk prediction pipeline using the UCI Cleveland Heart Disease dataset; trained and compared Logistic Regression and Random Forest classifiers, resolving overfitting via dataset cleaning and feature selection.
- Designed and deployed a fully offline Retrieval-Augmented Generation (RAG) chatbot using LangChain, ChromaDB, and Llama 3.2 via Ollama, ingesting AHA Hypertension and CHD clinical guidelines as a knowledge base.
- Built an agentic AI layer with five custom tools to intelligently route patient queries to the appropriate clinical guideline before generating responses.
- Integrated both ML and RAG backends into a FastAPI service and built a Streamlit web frontend combining prediction results with RAG chatbot responses in a single interface.
- Deployed the full system locally across coordinated multi-service backends (ML on port 8000, RAG agent on port 8001).
- *Tech stack: Python, Scikit-learn, LangChain, ChromaDB, Ollama, FastAPI, Streamlit*

## OTHER PROJECTS

### PashtoSenti: Transformer-Based Sentiment Analysis for Pashto (Research)

2026

- Authored a full research proposal for PashtoSenti, a first-of-its-kind benchmark dataset and NLP system for Pashto sentiment analysis — a severely under-resourced language with 60M+ speakers.
- Proposed pipeline covers social media data collection (Facebook, Twitter/X, YouTube), Pashto-specific preprocessing (script normalization, custom tokenization), and three-class sentiment annotation with inter-annotator agreement (Cohen's Kappa target  $\geq 0.70$ ).
- Benchmarking plan compares classical ML baselines (SVM, Naive Bayes, Logistic Regression) against multilingual transformers (mBERT, XLM-RoBERTa) and a PashtoCorp-adapted XLM-R; includes a cross-lingual Urdu-to-Pashto transfer learning experiment.
- Target publication: ACM TALLIP or IEEE Access; dataset and code planned for public release on HuggingFace and GitHub under CC-BY 4.0.
- *Tech stack: Python, HuggingFace Transformers, PyTorch, Scikit-learn, NLPashto, Label Studio*

### Frontend Development Intern – Youngdev Interns

Aug 2024

- Contributed to UI development for web applications using HTML, CSS, and JavaScript; participated in code reviews and agile planning sessions.

EDUCATION

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| <b>University of Peshawar</b><br><i>Bachelor of Science in Computer Science</i>                                                                                                                                                                                         | <i>Peshawar, Pakistan</i><br><i>Session 2022–2026 (Final Year)</i> |
| <ul style="list-style-type: none"><li>• Relevant coursework: Machine Learning, Neural Networks, Network Security, Cryptography, Data Structures, Algorithms</li><li>• Final Year Project: AI-Driven CVD Risk Prediction System (supervised by Dr. Rehman Ali)</li></ul> |                                                                    |
| <b>Hadaf Group of Colleges</b><br><i>Higher Secondary (Pre-Medical), Grade: A+</i>                                                                                                                                                                                      | <i>Peshawar, Pakistan</i><br><i>2019 – 2022</i>                    |
| <b>Islamia Collegiate School</b><br><i>Secondary Education (Science), Grade: A+</i>                                                                                                                                                                                     | <i>Peshawar, Pakistan</i><br><i>2015 – 2019</i>                    |

CERTIFICATIONS

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|-------------------------------------------------------------------------------------|-----------------------|
| • <b>Google UX Design Professional Certificate</b> – Coursera / Google              | <i>Jul – Sep 2024</i> |
| • <b>Google Soft Skills (Freelancing) Program</b> – PAFLA / Google                  | <i>Aug 2024</i>       |
| • <b>Data Science &amp; Analytics</b> – HP LIFE                                     | <i>Oct 2023</i>       |
| • <b>Object-Oriented Data Structures in C++</b> – University of Illinois (Coursera) | <i>Sep 2023</i>       |
| • <b>Introduction to C++ Programming</b> – Coursera Project Network                 | <i>Aug 2023</i>       |

LEADERSHIP & VOLUNTEERING

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| • <b>District Youth Cabinet Member</b> – Pakistan Youth Parliament (2024–25): Engaged in policy discussions and youth empowerment initiatives; bridging youth voices with decision-makers. |                           |
| • <b>Volunteer</b> – Regional Blood Center Peshawar                                                                                                                                        | <i>Aug 2022 – Present</i> |
| • <b>Volunteer</b> – Saylani Welfare International Trust: Organized community Iftar meals for laborers during Ramadan.                                                                     |                           |

LANGUAGES

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| <b>Pashto</b> – Native    — <b>Urdu</b> – Fluent    — <b>English</b> – Professional (B2) |
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